

Course Syllabus

Basic Information:

COLLAGE					
Collage Name	Collage of Engineering				
Department	Computer Engineering				
Semester	1 st Semester				
Year	2025/2026				
COURSE					
Course Name	Computer Organization & Assembly Language				
Course Code	ECOM 4412				
Credits	4				
Pre-requisite	ECOM 3421 Computer Architecture				
TEACHER					
Teacher Name	Ruba A. Salamah				
Office					
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Office Hours					
LECTURE	Building	Day	Start	End	Room Number
TEXTBOOK					
"Assembly Language for x86 Processors", by Kip R. Irvine (7th Edition), Prentice-Hall, 2015.					
REFERENACES					
"x86-64 Assembly Language Programming with Ubuntu ", by Ed Jorgensen, Ph.D. 2020. "Computer Organization And Design, the hardware / software interface", by David A. Patterson, and John L. Hennessy, 5th ed. 2014 "Introduction To Assembly Language Programming For Pentium and RISC Processors", by Sivarama P. Dandamudi (2nd Edition), SPringer 2005.					

Course Description:

This course introduces students to low-level programming using Assembly language. It provides a detailed understanding of how computer hardware executes instructions and manages data at the machine level. Topics include number systems, data representation, instruction formats, addressing modes, registers, memory organization, and the use of assembly instructions for arithmetic, logic, branching, and input/output operations. Students will gain hands-on experience in writing, assembling, linking, and debugging assembly programs. By the end of the course, students will understand the interaction between hardware and software and get full understanding of computer architecture and operating systems

Course Objectives:

Students who successfully complete this course will have demonstrated the ability to:

1. Understand the principles of computer architecture as applied to the Intel x86 processor family,
2. Understand data representation, including signed and unsigned integers, real numbers, and character data,
3. Apply the concepts of basic Boolean logic to computer hardware and programming,
4. Be familiar with the syntax and constructs of assembly language,
5. Create, compile, link and run assembly language programs using the Microsoft Visual Studio Integrated Development Environment,
6. Construct programs using one or more of the following: sequence, selection, and repetition,
7. Use procedures and arrays,
8. Solve problems and create solutions using assembly language.

Course Contents:

Schedule Table		
Week	Content	Notes
1	Basic Concepts: Applications of assembly language, basic concepts, machine language, and data representation.	
2	x86 Processor Architecture: Basic microcomputer design, instruction execution cycle, x86 processor architecture, Intel64 architecture, x86 memory management, components of a microcomputer, and the input–output system.	
3	Assembly Language Fundamentals: Introduction to assembly language, linking and debugging, and defining constants and variables.	
4	Data Transfers, Addressing, and Arithmetic: Data transfer and arithmetic instructions, assemble-link-execute	

5	cycle, operators, directives, expressions, JMP and LOOP instructions, and indirect addressing.	
6	Procedures: Linking to an external library, description of the book's link library, stack operations, defining and using procedures, flowcharts, and top-down structured design.	
7		
8	Conditional Processing: Boolean and comparison instructions, conditional jumps and loops, high-level logic structures, and finite-state machines.	
9		
10	Integer Arithmetic: Shift and rotate instructions with useful applications, multiplication and division, extended addition and subtraction, and ASCII and packed decimal arithmetic.	
11		
12	Advanced Procedures: Stack parameters, local variables, advanced PROC and INVOKE directives, and recursion.	
Self Read	Structures and Macros (Optional): Structures, macros, conditional assembly directives, and defining repeat blocks	
Self Read	MS-Windows Programming (Optional): Protected mode memory management concepts, using the Microsoft-Windows API to display text and colors, and dynamic memory allocation.	

Outcomes:

A. Knowledge

1. Students will demonstrate knowledge of basic Computer Organization: design logic; digital diagrams, and basic circuits and gates, and the link between Boolean functions, circuits, processor and machine code
2. Name and define the elements of an assembly language program.
3. Describe and express in an assembly language program, data structures used to solve common engineering, mathematical, and business-oriented problems.

B. Cognitive Skills

1. write, test and debug programs in x86 assembly language using assembler, debugger and emulation software
2. Employ assembly language directives and operators to effectively express code and data in an assembly language program.
3. Use current software engineering techniques to design and implement programs.
4. Employ structured techniques to construct branching and looping in assembly code.
5. Diagram and analyze the use of the hardware stack when designing procedures.

C. Interpersonal skills and responsibilities

1. Relate x86 assembly language with other processor assembly languages and high-level languages.
2. Able to understand the organization of different processors other than the x86 CPU.
3. Earn teamwork skills and display potential leadership qualities

D. Analysis and communication

1. Evaluate the efficiency of an assembly language program.
2. skills and motivation for independent learning and engagement in lifelong learning and research

Teaching Strategies:

- 1- Power point presentation Lecture
- 2- Problem Solving
- 3- Discussion with small groups
- 4- Learning by doing
- 5- Scientific Research strategy

Course activities:

1. Lab work
2. Research
3. Presentation
4. Programming problems

Assesment:

- 1- Labs, Activities and quizzes 30%
- 2- MidTerm Exam 30%
- 3- Final Exam 40%

Technology Used:

You need a computer that runs a 32-bit or 64-bit version of Microsoft Windows, along with one of the recent versions of Microsoft Visual Studio, as well as MASM assembler